Literature Review

StockWise AI – Retail Demand Forecasting & Sales Prediction

DEPI Machine Learning Capstone · Team of 5 · Version 1.1 · 31 October 2026

This document records the instructor's feedback on our project and how we responded to it. It also includes a short academic background for our main design choices, as the instructor requested. We update it after every review meeting.

Note: no model has been trained yet, so the numbers below are design choices or examples, not results.

Version history

Version	Date	Change
1.1	31 Oct 2026	Feedback from review meeting 1 and our responses.

1. Feedback & Evaluation

Meeting 1 (31 October 2026): the instructor reviewed our one-page project overview and gave the points below. He asked us to apply them to all documents we hand in, not only to the overview.

ID	Feedback	What we did	Where
F1-01	Explain why each tool was chosen, for example why XGBoost.	Added the reason for each model and tool.	This doc §2.1; Planning §1.6; Requirements FR-19
F1-02	State how many days the lag variables use.	Fixed the lags at 16, 21 and 28 days.	This doc §2.2; Planning §1.3; Requirements FR-12, FR-13
F1-03	How will the AI assistant avoid hallucination?	Designed the assistant so all numbers come from our own data.	This doc §2.3; Planning risk R12; Requirements FR-36 – FR-42
F1-04	How are safety stock and reorder point calculated in the simulation?	Wrote the formulas and the simulation steps.	This doc §2.4; Planning risk R13; Requirements FR-29 – FR-31, FR-35
F1-05	The overview is too technical; add an academic part.	Added a short academic background with references.	This doc §3
F1-06	Apply the feedback to all handed-in documents.	Updated Planning and Requirements to version 1.1.	Both documents, “Changes in Version 1.1”

2. Our Responses

2.1 Why we chose these models (F1-01)

Model	Why
Naive baseline	Predicts the sales of the same weekday three weeks earlier. Any real model must beat it.
Linear Regression	Required by the brief. Simple and easy to explain, so it is our reference point.
Polynomial Regression	Required by the brief. Tests whether curved relationships improve on the straight line.
Random Forest	Many decision trees averaged together. Handles non-linear patterns with little tuning.
XGBoost	Our main candidate (see below).

We expect XGBoost to do well for three reasons. Tree-boosting models were among the best methods in the M5 competition on Walmart sales (Makridakis et al., 2022). Our data is a table of features, where tree models usually work well (Grinsztajn et al., 2022). It is also fast on a normal CPU and shows which features matter most (Chen & Guestrin, 2016). We will only keep it if it beats the other models on the validation period.

We left out LSTM because it is not required, needs more computing power and tuning, and is harder to explain for a first ML project. The reason for each tool (Colab, scikit-learn, MLflow, GitHub, Streamlit) is listed in the Planning document, §1.6.

2.2 Lag days (F1-02)

A lag is what the same store sold of the same product family N days earlier. We forecast 16 days ahead, so on the last forecast day the newest sales we know are 16 days old. That means no lag can be shorter than 16 days, otherwise the model would use information it will not have in real life (data leakage).

Feature	Days back	Why
lag_16	16	The newest value we always know.
lag_21	21	Same weekday 3 weeks ago (keeps the weekly pattern). Also our naive baseline.
lag_28	28	Same weekday 4 weeks ago.
roll_mean_7	16 – 22	Average of one week, ending 16 days back.
roll_mean_28	16 – 43	Average of about one month, ending 16 days back.

Example: to predict 15 August 2017, lag_16 uses 30 July, which is the last day we know. All lags are calculated separately for each store and product family. We may test other lags later and keep them only if they lower the validation error.

2.3 Preventing AI hallucination (F1-03)

Hallucination is when an AI model gives an answer that sounds right but is made up (Ji et al., 2023). Our rule is that the AI never produces numbers itself. It only picks one of our functions, our code reads the answer from our forecast tables, and the AI explains it in plain words. The guardrails are:

- G1: all numbers come from our functions (for example get_forecast), not from the AI.
- G2: functions accept only real stores (1–54), the 33 product families and dates in the forecast window.
- G3: the AI is told to say “I don’t have that data” instead of guessing.
- G4: each answer shows the store, product family, dates and model it used.

- G5: code checks that every number in the answer matches the table before showing it.
- G6: temperature is set to 0 so answers stay consistent.
- G7: the assistant only recommends; a person makes every ordering decision.

We will test it with 10 prepared questions, including trick questions such as a store that does not exist. The target is at least 8 of 10 correct and all trick questions refused.

2.4 Safety stock and reorder point in the simulation (F1-04)

The dataset has sales but no stock levels or delivery times, so we simulate those. The forecasts and their errors come from our real model. We use the standard formulas (Silver et al., 2017):

- Safety stock = $z \times \sigma \times \sqrt{L}$
- Reorder point = $D \times L$ + safety stock
- Order when the stock we have plus stock on order is at or below the reorder point.

Here D is the average daily forecast, L is the lead time in days, σ is the spread of our model's daily errors on the validation period, and $z = 1.65$ for a 95% chance of not running out. Better forecasts give a smaller σ, so less safety stock is needed.

Example (made-up numbers): D = 20 units a day, $\sigma = 6$, L = 4 days. Safety stock = $1.65 \times 6 \times 2 \approx 20$ units. Reorder point = $20 \times 4 + 20 = 100$ units. With 85 units in stock, the system recommends ordering.

The simulation runs over the 16 test days using the real sales. We run it once with the naive forecast and once with our best model, and compare stockout days and average stock. Proposed values to confirm with the instructor: lead time 4 days (also testing 2 and 7), 95% service level, orders covering 7 days.

3. Academic Background (F1-05)

3.1 Retail demand forecasting

Supermarkets need daily forecasts for each product in each store to plan orders. This is hard because there are many series, many slow-selling items, and strong effects from promotions and holidays. The Kaggle Favorita dataset we use has all of these: 54 stores, 33 product families and daily sales from 2013 to 2017.

3.2 What forecasting competitions show

In the M5 competition, teams forecast 42,840 Walmart sales series. Gradient-boosted tree models were among the top methods, and using extra information such as calendar events was important (Makridakis et al., 2022). This supports our use of XGBoost and of calendar, promotion and holiday features.

3.3 Models for table data

Grinsztajn et al. (2022) compared tree models with deep learning on many table datasets and found that tree models often perform better. Their datasets were smaller than ours, but together with the M5 results this is why we start with tree models and leave LSTM out of this version.

3.4 Evaluating forecasts

A forecast must be tested on dates after the training data, never on shuffled rows (Hyndman & Athanasopoulos, 2021). We also avoid MAPE, because it breaks when actual sales are zero (Hyndman & Koehler, 2006), which happens often in our data. We use MAE, RMSE and WAPE, and always compare with the naive baseline.

4. Final Grading Criteria

The DEPI guidelines split the grade between documentation, implementation, testing and presentation. We have not received the weights yet.

5. References

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794).

Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems 35*.

Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts.

Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting, 22*(4), 679–688.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys, 55*(12), Article 248.

Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting, 38*(4), 1346–1364.

Silver, E. A., Pyke, D. F., & Thomas, D. J. (2017). *Inventory and production management in supply chains* (4th ed.). CRC Press.